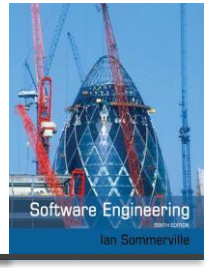
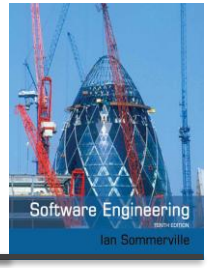


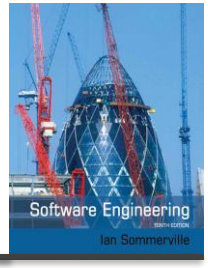
Chapter 18 – Service-oriented Software Engineering

Topics covered



- ✧ Service-oriented architectures
- ✧ RESTful services
- ✧ Service engineering
- ✧ Service composition

Web services



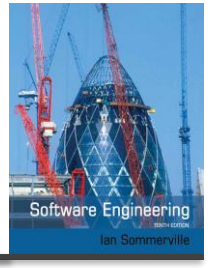
- A web service is an instance of a more general notion of a service:

  "an act or performance offered by one party to another. Although the process may be tied to a physical product, the performance is essentially intangible and does not normally result in ownership of any of the factors of production".

- The essence of a service, therefore, is that the provision of the service is independent of the application using the service.

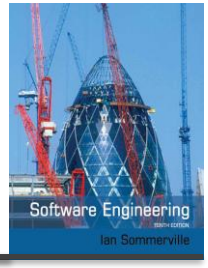
- Service providers can develop specialized services and offer these to a range of service users from different organizations.

Reusable services



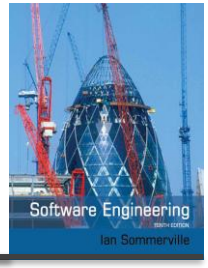
- Services are reusable components that are independent (no requires interface) and are loosely coupled.
- A web service is:
  - *A loosely coupled, reusable software component that encapsulates discrete functionality, which may be distributed and programmatically accessed. A web service is a service that is accessed using standard Internet and XML-based protocols.*
- Services are platform and implementation-language independent

Benefits of service-oriented apprach



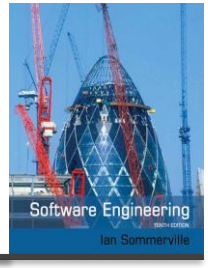
- Services can be offered by any service provider inside or outside of an organisation so organizations can create applications by integrating services from a range of providers.
- The service provider makes information about the service public so that any authorised user can use the service.
- Applications can delay the binding of services until they are deployed or until execution. This means that applications can be reactive and adapt their operation to cope with changes to their execution environment.

Benefits of a service-oriented approach



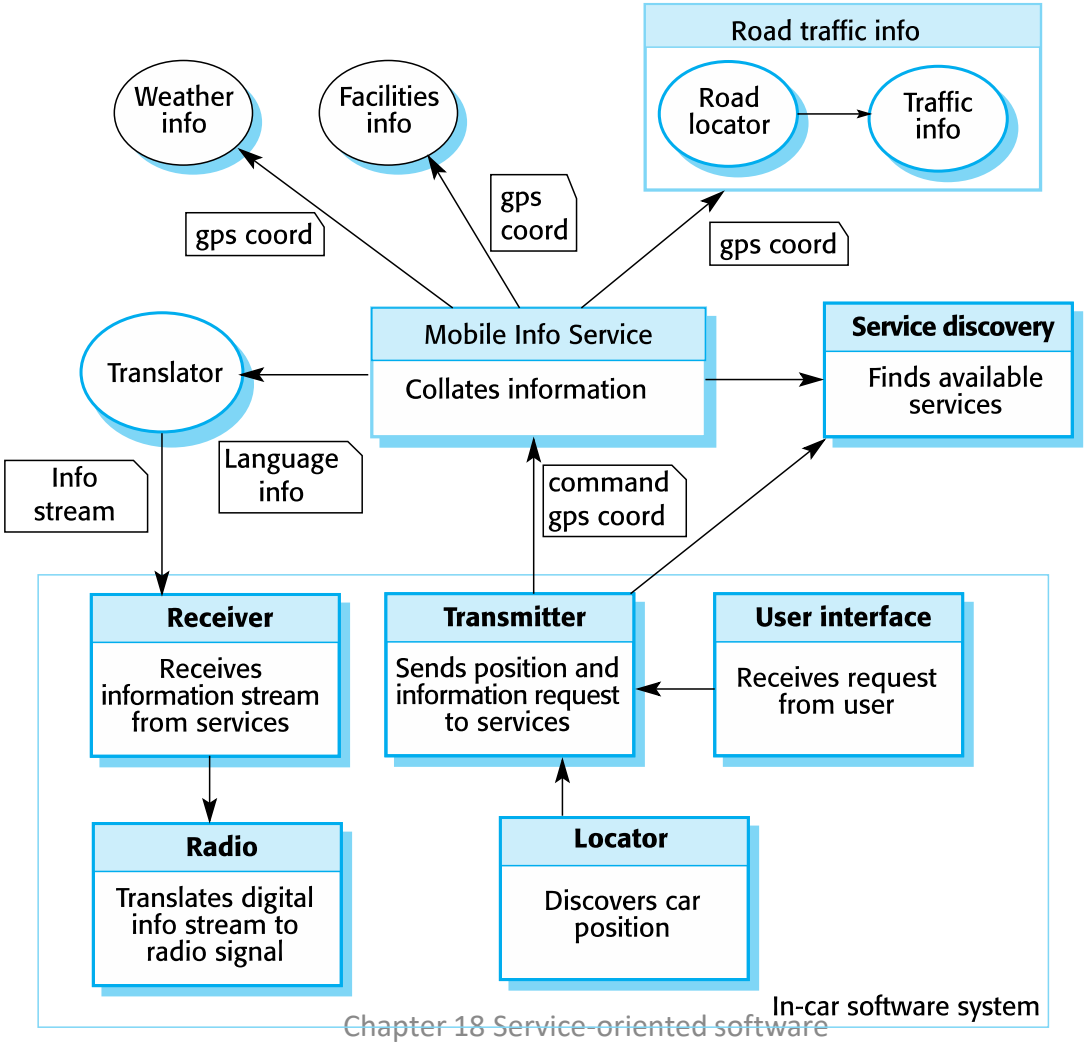
- Opportunistic construction of new services is possible. A service provider may recognise new services that can be created by linking existing services in innovative ways.
- Service users can pay for services according to their use rather than their provision. Instead of buying a rarely-used component, the application developers can use an external service that will be paid for only when required.
- Applications can be made smaller, which is particularly important for mobile devices with limited processing and memory capabilities. Computationally-intensive processing can be offloaded to external services.

Services scenario

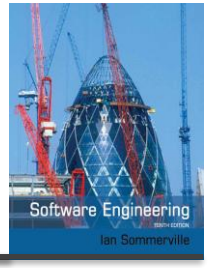


- An in-car information system provides drivers with information on weather, road traffic conditions, local information etc. This is linked to car audio system so that information is delivered as a signal on a specific channel.
- The car is equipped with GPS receiver to discover its position and, based on that position, the system accesses a range of information services. Information may be delivered in the driver's specified language.

A service-based, in-car information system

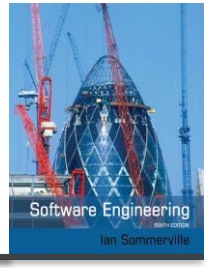


Advantage of SOA for this application

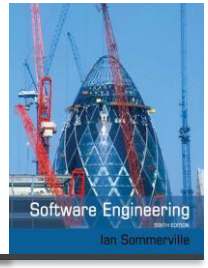


- It is not necessary to decide when the system is programmed or deployed what service provider should be used or what specific services should be accessed.
  - As the car moves around, the in-car software uses the service discovery service to find the most appropriate information service and binds to that.
  - Because of the use of a translation service, it can move across borders and therefore make local information available to people who don't speak the local language.

Service-oriented software engineering

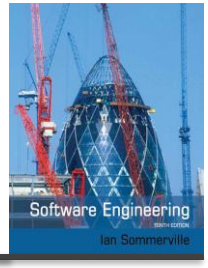


- As significant a development as object-oriented development.
- Building applications based on services allows companies and other organizations to cooperate and make use of each other's business functions.
- Service-based applications may be constructed by linking services from various providers using either a standard programming language or a specialized workflow language.



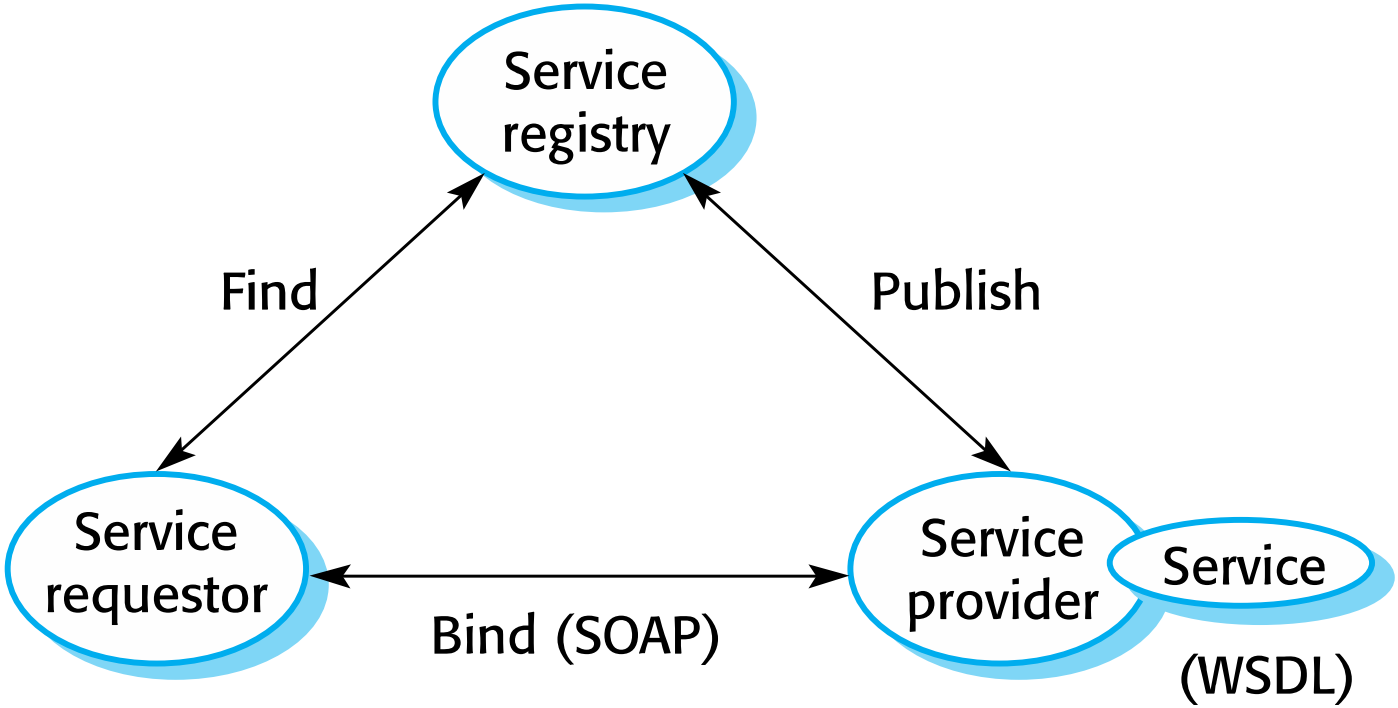
Service-oriented architecture

Service-oriented architectures

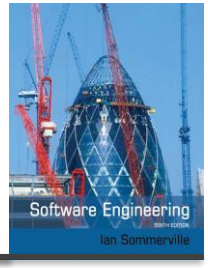


- A means of developing distributed systems where the components are stand-alone services
- Services may execute on different computers from different service providers
- Standard protocols have been developed to support service communication and information exchange

Service-oriented architecture

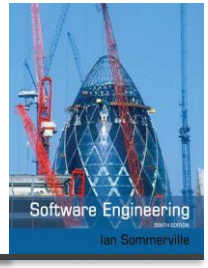


Benefits of SOA



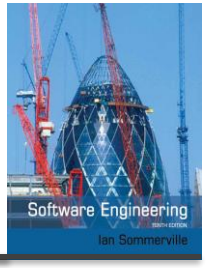
- Services can be provided locally or outsourced to external providers
- Services are language-independent
- Investment in legacy systems can be preserved
- Inter-organisational computing is facilitated through simplified information exchange

Key standards



- SOAP
  - A message exchange standard that supports service communication
- WSDL (Web Service Definition Language)
  - This standard allows a service interface and its bindings to be defined
- WS-BPEL
  - A standard for workflow languages used to define service composition

Web service standards



XML technologies (XML, XSD, XSLT, ...)

Support (WS-Security, WS-Addressing, ...)

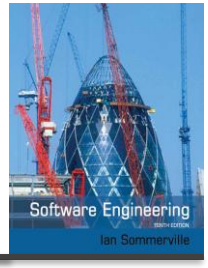
Process (WS-BPEL)

Service definition (UDDI, WSDL)

Messaging (SOAP)

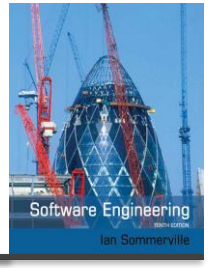
Transport (HTTP, HTTPS, SMTP, ...)

Service-oriented software engineering



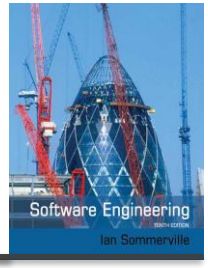
- Existing approaches to software engineering have to evolve to reflect the service-oriented approach to software development
  - Service engineering. The development of dependable, reusable services
    - Software development for reuse
  - Software development with services. The development of dependable software where services are the fundamental components
    - Software development with reuse

Services as reusable components



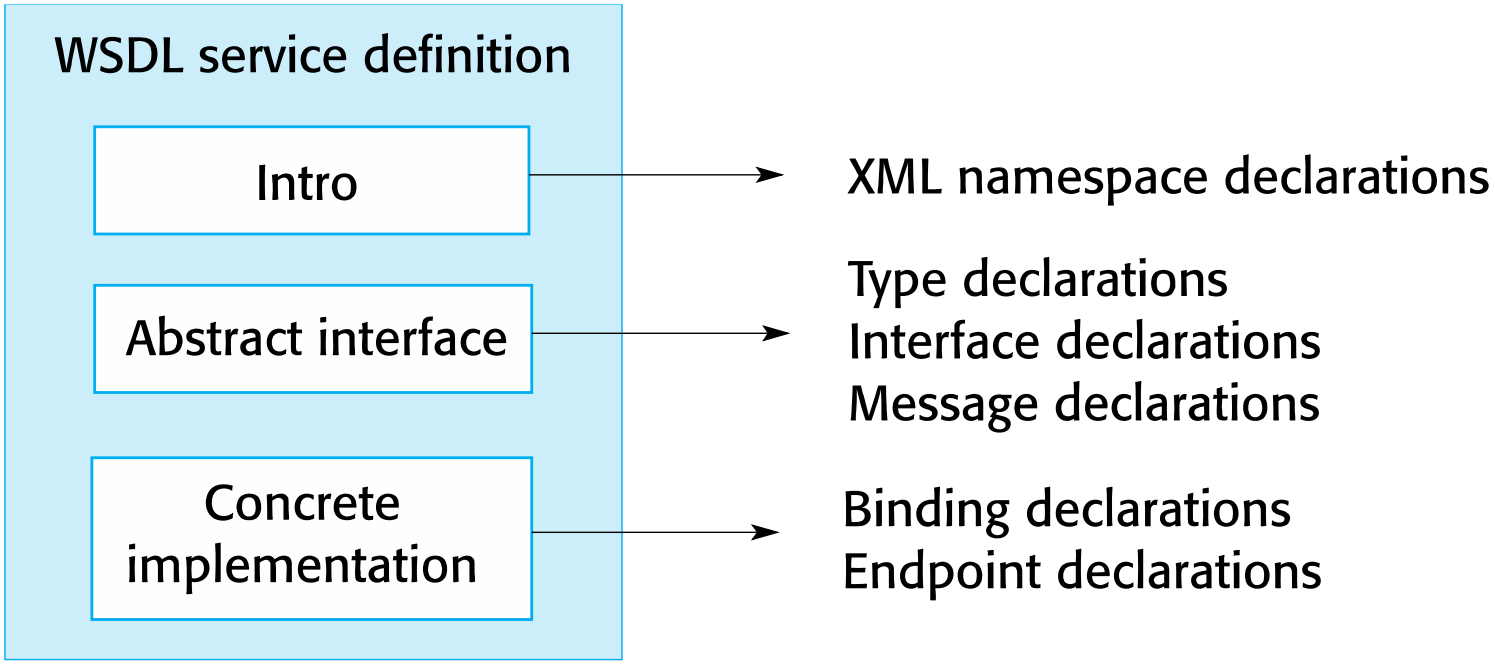
- A service can be defined as:
  - *A loosely-coupled, reusable software component that encapsulates discrete functionality which may be distributed and programmatically accessed. A web service is a service that is accessed using standard Internet and XML-based protocols*
- A critical distinction between a service and a component as defined in CBSE is that services are independent
  - Services do not have a 'requires' interface
  - Services rely on message-based communication with messages expressed in XML

Web service description language

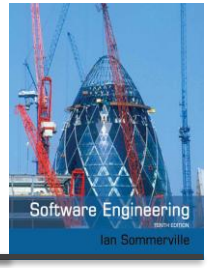


- The service interface is defined in a service description expressed in WSDL (Web Service Description Language).
- The WSDL specification defines
  - What operations the service supports and the format of the messages that are sent and received by the service
  - How the service is accessed - that is, the binding maps the abstract interface onto a concrete set of protocols
  - Where the service is located. This is usually expressed as a URI (Universal Resource Identifier)

Organization of a WSDL specification

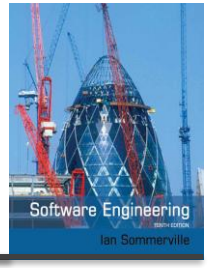


WSDL specification components



- The ‘what’ part of a WSDL document, called an interface, specifies what operations the service supports, and defines the format of the messages that are sent and received by the service.
- The ‘how’ part of a WSDL document, called a binding, maps the abstract interface to a concrete set of protocols. The binding specifies the technical details of how to communicate with a Web service.
- The ‘where’ part of a WSDL document describes the location of a specific Web service implementation (its endpoint).

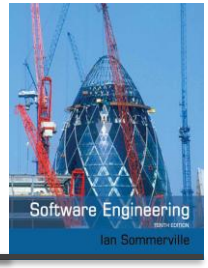
Part of a WSDL description for a web service



Define some of the types used. Assume that the namespace prefixes 'ws' refers to the namespace URI for XML schemas and the namespace prefix associated with this definition is weathns.

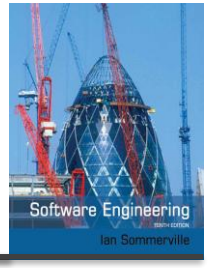
```
<types>
  <xs: schema targetNameSpace = "http://.../weathns"
    xmlns: weathns = "http://…/weathns" >
    <xs:element name = "PlaceAndDate" type = "pdrec" />
    <xs:element name = "MaxMinTemp" type = "mmtrec" />
    <xs: element name = "InDataFault" type = "errmess" />

    <xs: complexType name = "pdrec"
    <xs: sequence>
    <xs:element name = "town" type = "xs:string"/>
    <xs:element name = "country" type = "xs:string"/>
    <xs:element name = "day" type = "xs:date" />
    </xs:complexType>
    Definitions of MaxMinType and InDataFault here
  </schema>
</types>
```

Part of a WSDL description for a web service



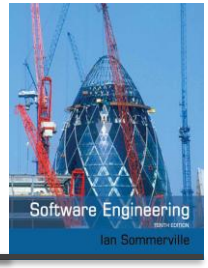
Now define the interface and its operations. In this case, there is only a single operation to return maximum and minimum temperatures.

```
<interface name = “weatherInfo” >
  <operation name = “getMaxMinTemps” pattern = “wsdlns: in-out”>
    <input messageLabel = “In” element = “weathns: PlaceAndDate” />
    <output messageLabel = “Out” element = “weathns:MaxMinTemp” />
    <outfault messageLabel = “Out” element = “weathns:InDataFault” />
  </operation>
</interface>
```



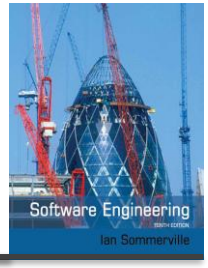
RESTful services

RESTful web services



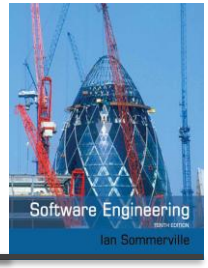
- Current web services standards have been criticized as ‘heavyweight’ standards that are over-general and inefficient.
- REST (REpresentational State Transfer) is an architectural style based on transferring representations of resources from a server to a client.
- This style underlies the web as a whole and is simpler than SOAP/WSDL for implementing web services.
- RESTful services involve a lower overhead than so-called ‘big web services’ and are used by many organizations implementing service-based systems.

Resources



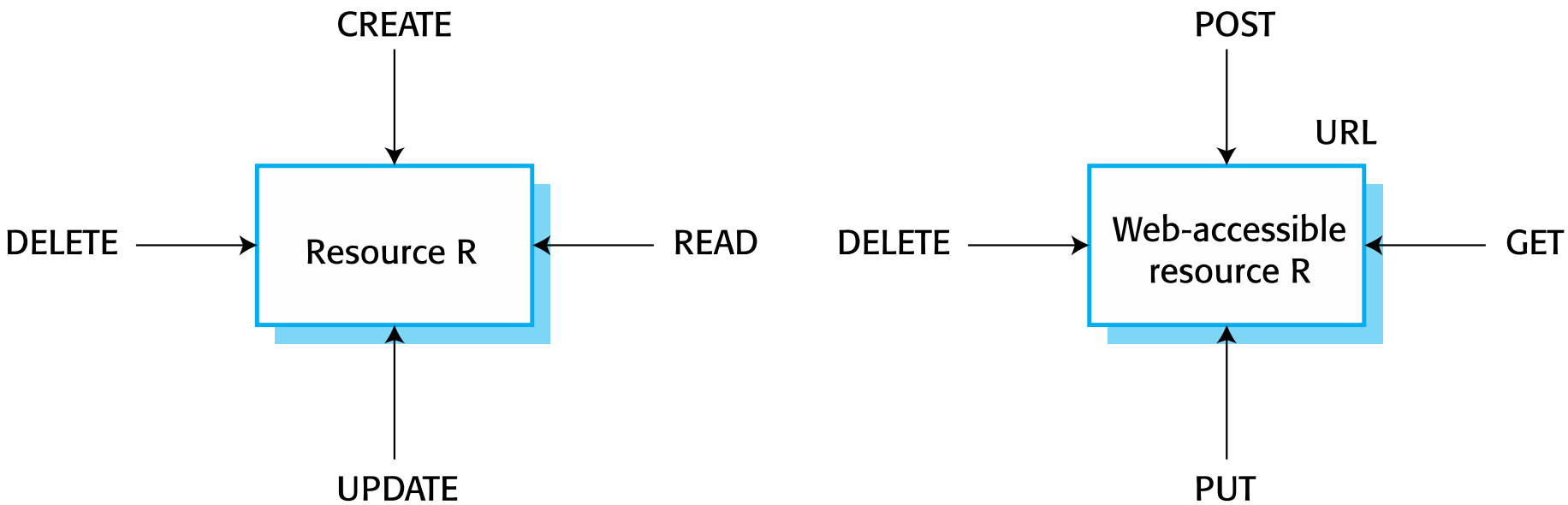
- The fundamental element in a RESTful architecture is a resource.
- Essentially, a resource is simply a data element such as a catalog, a medical record, or a document, such as this book chapter.
- In general, resources may have multiple representations i.e. they can exist in different formats.
  - MS WORD
  - PDF
  - Quark XPress

Resource operations



- ✧ Create – bring the resource into existence.
- ✧ Read – return a representation of the resource.
- ✧ Update – change the value of the resource.
- ✧ Delete – make the resource inaccessible.

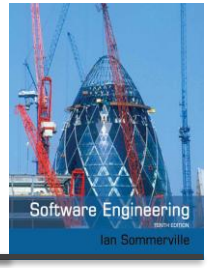
Resources and actions



(a) General resource actions

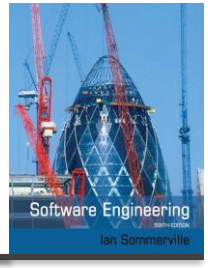
(b) Web resources

Operation functionality



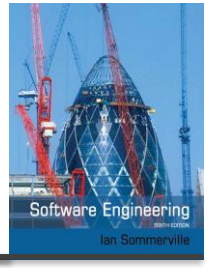
- ✧ POST is used to create a resource. It has associated data that defines the resource.
- ✧ GET is used to read the value of a resource and return that to the requestor in the specified representation, such as XHTML, that can be rendered in a web browser.
- ✧ PUT is used to update the value of a resource.
- ✧ DELETE is used to delete the resource.

Resource access



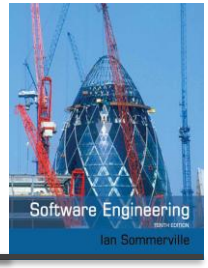
- When a RESTful approach is used, the data is exposed and is accessed using its URL.
- Therefore, the weather data for each place in the database, might be accessed using URLs such as:
  - http://weather-info-example.net/temperatures/boston http://weather-info-example.net/temperatures/edinburgh
- Invokes the GET operation and returns a list of maximum and minimum temperatures.
- To request the temperatures for a specific date, a URL query is used:
  - http://weather-info-example.net/temperatures/edinburgh?date=20140226

Query results



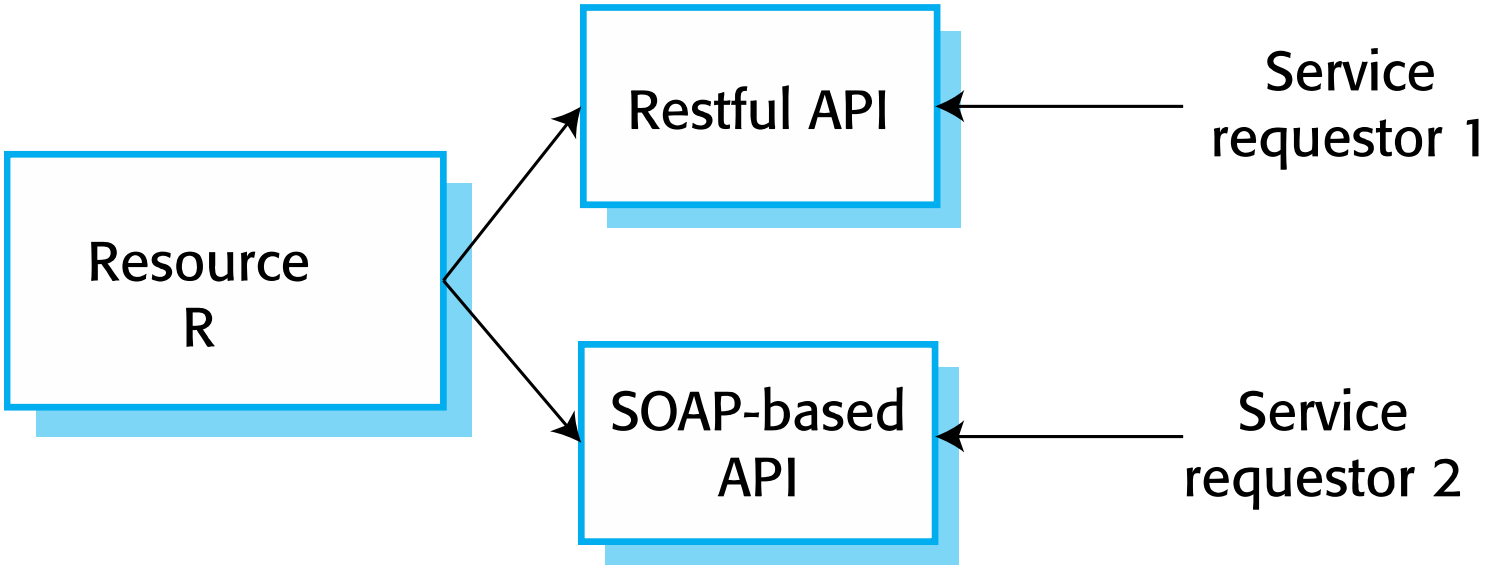
- The response to a GET request in a RESTful service may include URLs.
- If the response to a request is a set of resources, then the URL of each of these may be included.
  - http://weather-info-example.net/temperatures/edinburgh-scotland http://weather-info-example.net/temperatures/edinburgh-australia http://weather-info-example.net/temperatures/edinburgh-maryland

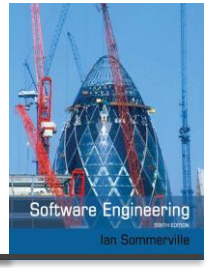
Disadvantages of RESTful approach



- When a service has a complex interface and is not a simple resource, it can be difficult to design a set of RESTful services to represent this.
- There are no standards for RESTful interface description so service users must rely on informal documentation to understand the interface.
- When you use RESTful services, you have to implement your own infrastructure for monitoring and managing the quality of service and the service reliability.

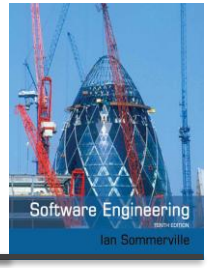
RESTful and SOAP-based APIs





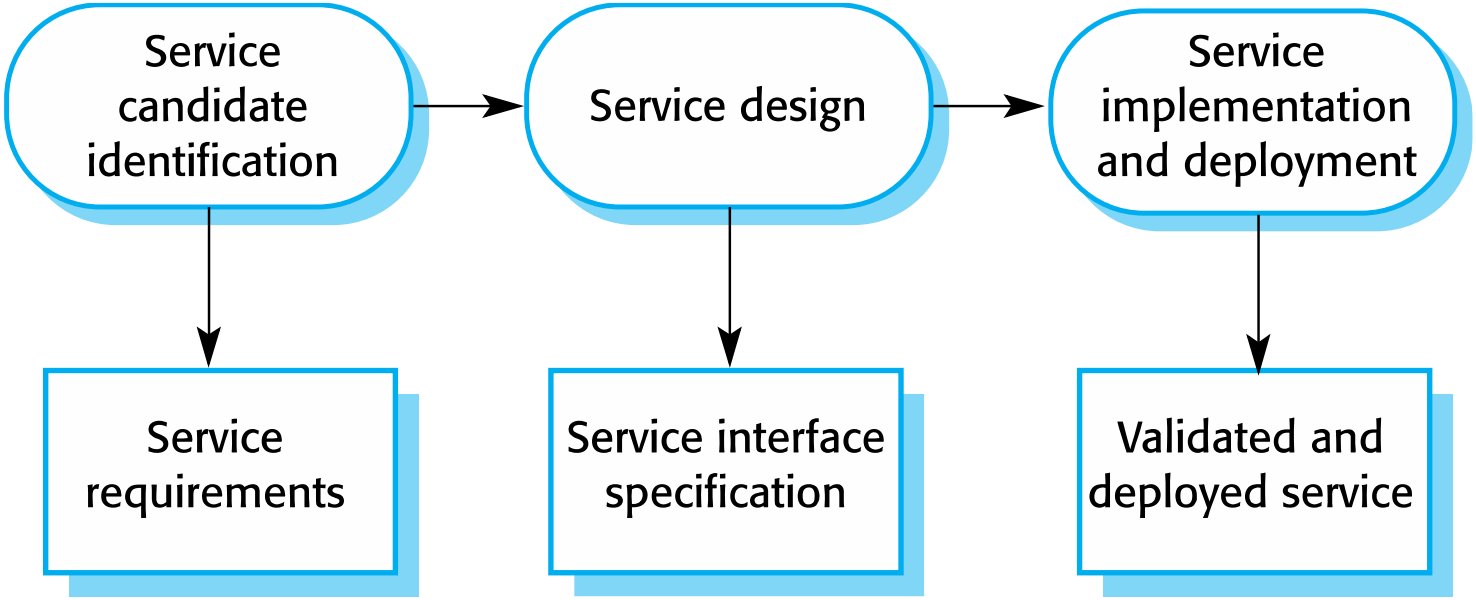
Service engineering

Service engineering

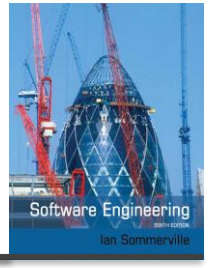


- The process of developing services for reuse in service-oriented applications
- The service has to be designed as a reusable abstraction that can be used in different systems.
- Generally useful functionality associated with that abstraction must be designed and the service must be robust and reliable.
- The service must be documented so that it can be discovered and understood by potential users.

The service engineering process

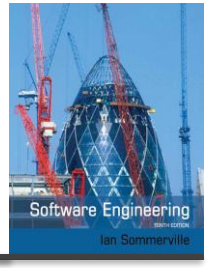


Stages of service engineering



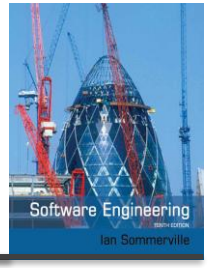
- Service candidate identification, where you identify possible services that might be implemented and define the service requirements.
- Service design, where you design the logical service interface and its implementation interfaces (SOAP and/or RESTful)
- Service implementation and deployment, where you implement and test the service and make it available for use.

Service candidate identification



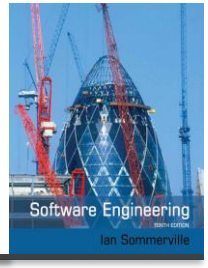
- Services should support business processes.
- Service candidate identification involves understanding an organization's business processes to decide which reusable services could support these processes.
- Three fundamental types of service
  - Utility services that implement general functionality used by different business processes.
  - Business services that are associated with a specific business function e.g., in a university, student registration.
  - Coordination services that support composite processes such as ordering.

Task and entity-oriented services



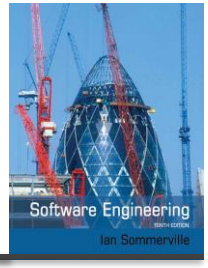
- Task-oriented services are those associated with some activity.
- Entity-oriented services are like objects. They are associated with a business entity such as a job application form.
- Utility or business services may be entity- or task-oriented, coordination services are always task-oriented.

Service classification



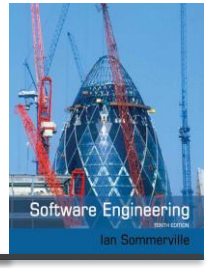
	Utility	Business	Coordination
Task	Currency converter Employee locator	Validate claim form Check credit rating	Process expense claim Pay external supplier
Entity	Document style checker Web form to XML converter	Expenses form Student application form	

Service identification



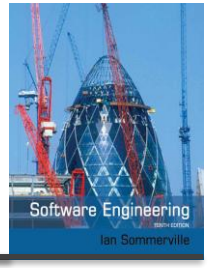
- Is the service associated with a single logical entity used in different business processes?
- Is the task one that is carried out by different people in the organisation? Can this fit with a RESTful model?
- Is the service independent?
- Does the service have to maintain state? Is a database required?
- Could the service be used by clients outside the organisation?
- Are different users of the service likely to have different non-functional requirements?

Service identification example



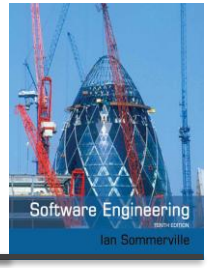
- A large company, which sells computer equipment, has arranged special prices for approved configurations for some customers.
- To facilitate automated ordering, the company wishes to produce a catalog service that will allow customers to select the equipment that they need.
- Unlike a consumer catalog, orders are not placed directly through a catalog interface. Instead, goods are ordered through the web-based procurement system of each company that accesses the catalog as a web service.
- Most companies have their own budgeting and approval procedures for orders and their own ordering process must be followed when an order is placed.

Catalog services



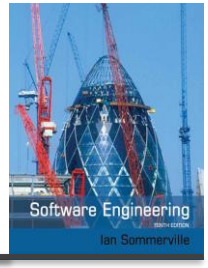
- Created by a supplier to show which good can be ordered from them by other companies
- Service requirements
  - Specific version of catalogue should be created for each client
  - Catalogue shall be downloadable
  - The specification and prices of up to 6 items may be compared
  - Browsing and searching facilities shall be provided
  - A function shall be provided that allows the delivery date for ordered items to be predicted
  - Virtual orders shall be supported which reserve the goods for 48 hours to allow a company order to be placed

Catalogue: Non-functional requirements



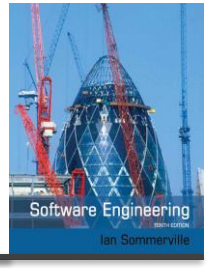
- Access shall be restricted to employees of accredited organisations
- Prices and configurations offered to each organisation shall be confidential
- The catalogue shall be available from 0700 to 1100
- The catalogue shall be able to process up to 10 requests per second

Functional descriptions of catalog service operations



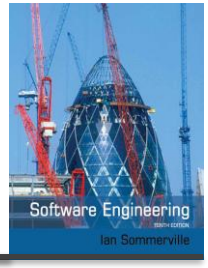
Operation	Description
MakeCatalog	Creates a version of the catalog tailored for a specific customer. Includes an optional parameter to create a downloadable PDF version of the catalog.
Lookup	Displays all of the data associated with a specified catalog item.
Search	This operation takes a logical expression and searches the catalog according to that expression. It displays a list of all items that match the search expression.

Functional descriptions of catalog service operations



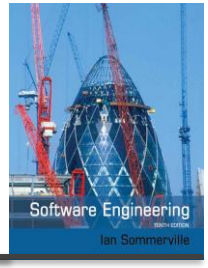
Operation	Description
Compare	Provides a comparison of up to six characteristics (e.g., price, dimensions, processor speed, etc.) of up to four catalog items.
CheckDelivery	Returns the predicted delivery date for an item if ordered that day.
MakeVirtualOrder	Reserves the number of items to be ordered by a customer and provides item information for the customer's own procurement system.

Service interface design



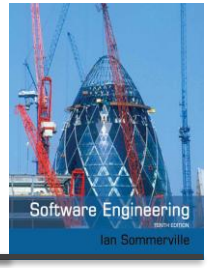
- Involves thinking about the operations associated with the service and the messages exchanged
- The number of messages exchanged to complete a service request should normally be minimised.
- Service state information may have to be included in messages

Interface design stages



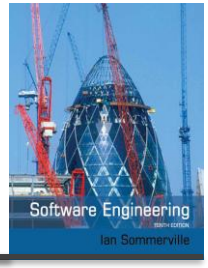
- Logical interface design
  - Starts with the service requirements and defines the operation names and parameters associated with the service. Exceptions should also be defined
- Message design (SOAP)
  - For SOAP-based services, design the structure and organisation of the input and output messages. Notations such as the UML are a more abstract representation than XML
  - The logical specification is converted to a WSDL description
- Interface design (REST)
  - Design how the required operations map onto REST operations and what resources are required.

Catalog interface design



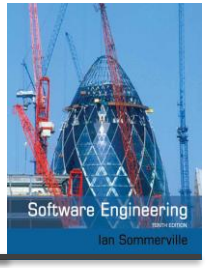
Operation	Inputs	Outputs	Exceptions
MakeCatalog	*mcIn* Company id PDF-flag	*mcOut* URL of the catalog for that company	*mcFault* Invalid company id
Lookup	*lookIn* Catalog URL Catalog number	*lookOut* URL of page with the item information	*lookFault* Invalid catalog number
Search	*searchIn* Catalog URL Search string	*searchOut* URL of web page with search results	*searchFault* Badly formed search string

Catalog interface design



Operation	Inputs	Outputs	Exceptions
Compare	*compIn* Catalog URL Entry attribute (up to 6) Catalog number (up to 4)	*compOut* URL of page showing comparison table	*compFault* Invalid company id Invalid catalog number Unknown attribute
CheckDelivery	*cdIn* Company id Catalog number Number of items required	*cdOut* Catalog number Expected delivery date	*cdFault* Invalid company id No availability Zero items requested
MakeVirtualOrder	*poIn* Company id Number of items required Catalog number	*poOut* Catalog number Number of items required Predicted delivery date Unit price estimate Total price estimate	*poFault* Invalid company id Invalid catalog number Zero items requested

UML definition of input and output messages



cdIn

cID: string
catNum: string
numItems: integer

size (cID) = 6
size (catNum) = 10
numItems > 0

cdOut

catNum: string
delivDate: date

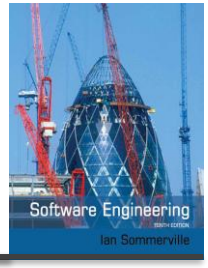
size (catNum) = 10
delivDate > Today

cdFault

errCode: integer

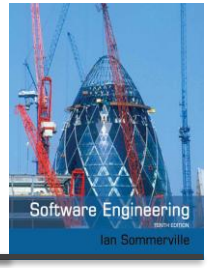
Invalid company id
errCode=1
Invalid catalog number
errCode = 2
No availability
errCode = 3
Zero items requested
errCode = 4

RESTful interface



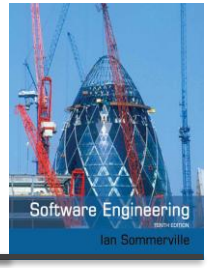
- There should be a resource representing a company-specific catalog. This should have a URL of the form <base catalog>/<company name> and should be created using a POST operation.
- Each catalog item should have its own URL of the form:
  - <base catalog>/<company name>/<item identifier>.
- The GET operation is used to retrieve items.
  - **Lookup** is implemented by using the URL of an item in a catalog as the GET parameter.
  - **Search** is implemented by using GET with the company catalog as the URL and the search string as a query parameter. This GET operation returns a list of URLs of the items matching the search.

RESTful interface



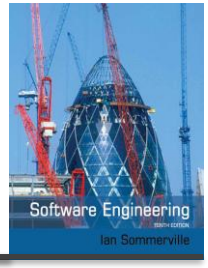
- The **Compare** operation can be implemented as a sequence of GET operations, to retrieve the individual items, followed by a POST operation to create the comparison table and a final GET operation to return this to the user.
- The **CheckDelivery** and **MakeVirtualOrder** operations require an additional resource, representing a virtual order.
  - A POST operation is used to create this resource with the number of items required. The company id is used to automatically fill in the order form and the delivery date is calculated. This can then be retrieved using a GET operation.

Service implementation and deployment



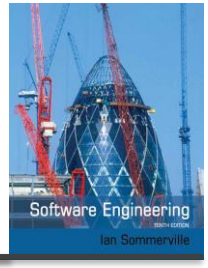
- Programming services using a standard programming language or a workflow language
- Services then have to be tested by creating input messages and checking that the output messages produced are as expected
- Deployment involves publicising the service and installing it on a web server. Current servers provide support for service installation

Legacy system services

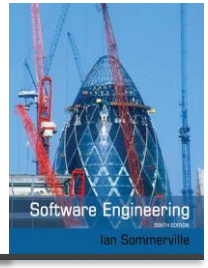


- Services can be implemented by implementing a service interface to existing legacy systems
- Legacy systems offer extensive functionality and this can reduce the cost of service implementation
- External applications can access this functionality through the service interfaces

Service descriptions

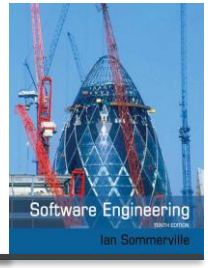


- Information about your business, contact details, etc. This is important for trust reasons. Users of a service have to be confident that it will not behave maliciously.
- An informal description of the functionality provided by the service. This helps potential users to decide if the service is what they want.
- A description of how to use the services SOAP-based and RESTful.
- Subscription information that allows users to register for information about updates to the service.



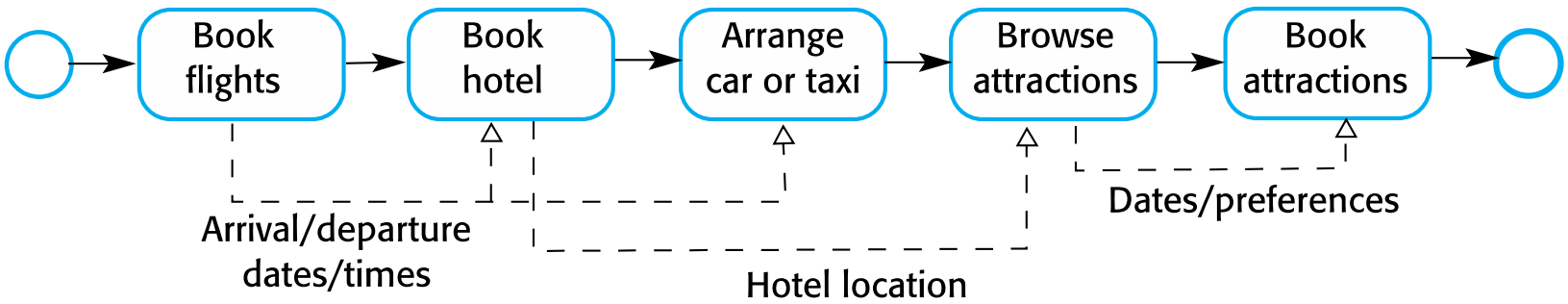
Service composition

Software development with services

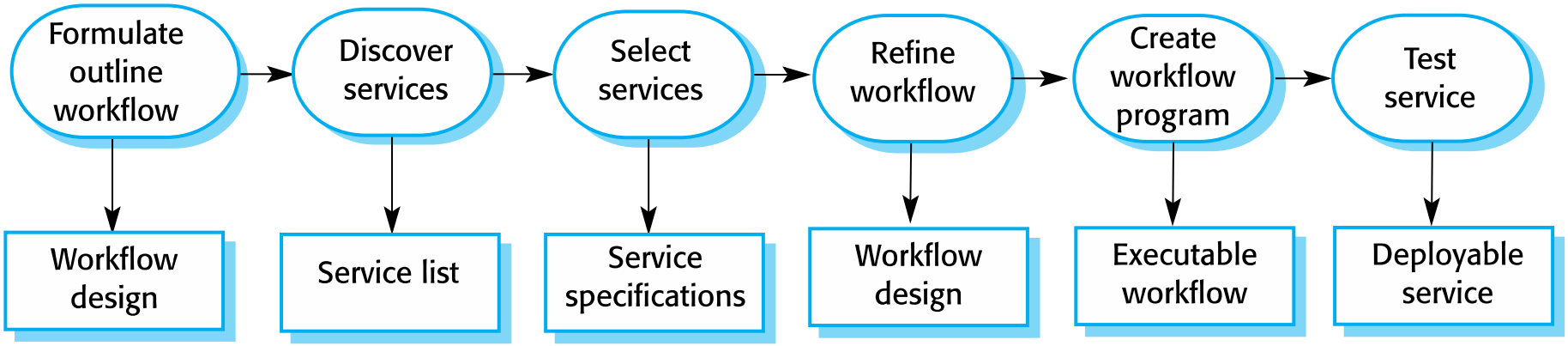


- Existing services are composed and configured to create new composite services and applications
- The basis for service composition is often a workflow
  - Workflows are logical sequences of activities that, together, model a coherent business process
  - For example, provide a travel reservation services which allows flights, car hire and hotel bookings to be coordinated

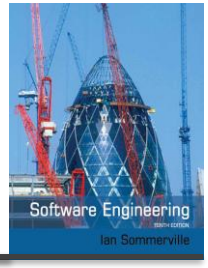
Vacation package workflow



Service construction by composition

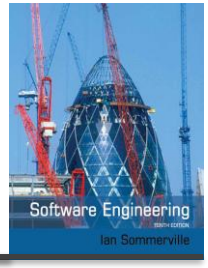


Construction by composition



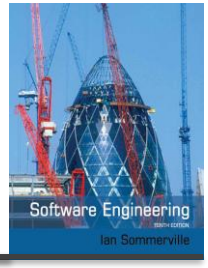
- *Formulate outline workflow*
  - In this initial stage of service design, you use the requirements for the composite service as a basis for creating an 'ideal' service design.
- *Discover services*
  - During this stage of the process, you search service registries or catalogs to discover what services exist, who provides these services and the details of the service provision.
- *Select possible services*
  - Your selection criteria will obviously include the functionality of the services offered. They may also include the cost of the services and the quality of service (responsiveness, availability, etc.) offered.

Construction by composition



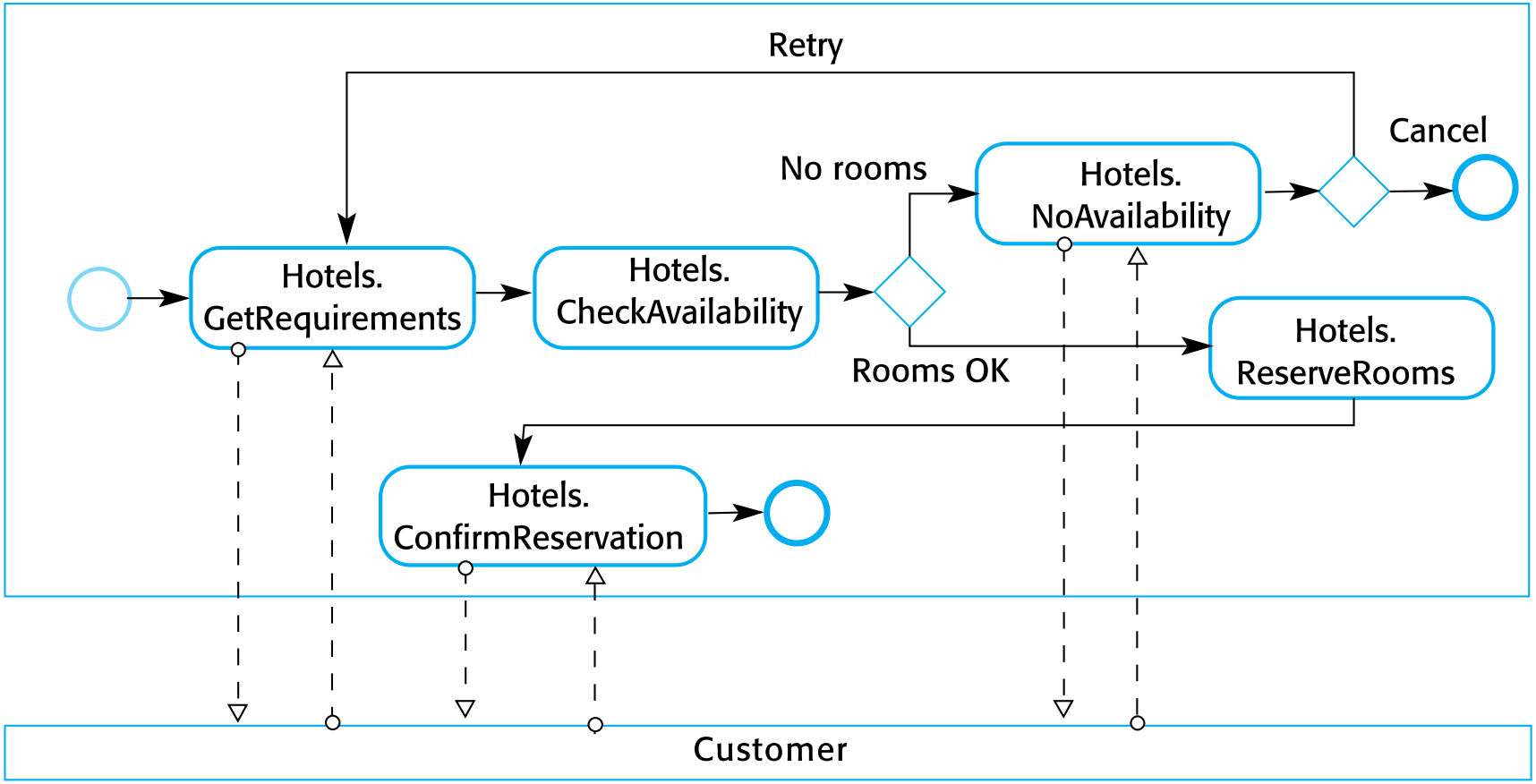
- *Refine workflow*.
  - This involves adding detail to the abstract description and perhaps adding or removing workflow activities.
- *Create workflow program*
  - During this stage, the abstract workflow design is transformed to an executable program and the service interface is defined. You can use a conventional programming language, such as Java or a workflow language, such as WS-BPEL.
- *Test completed service or application*
  - The process of testing the completed, composite service is more complex than component testing in situations where external services are used.

Workflow design and implementation

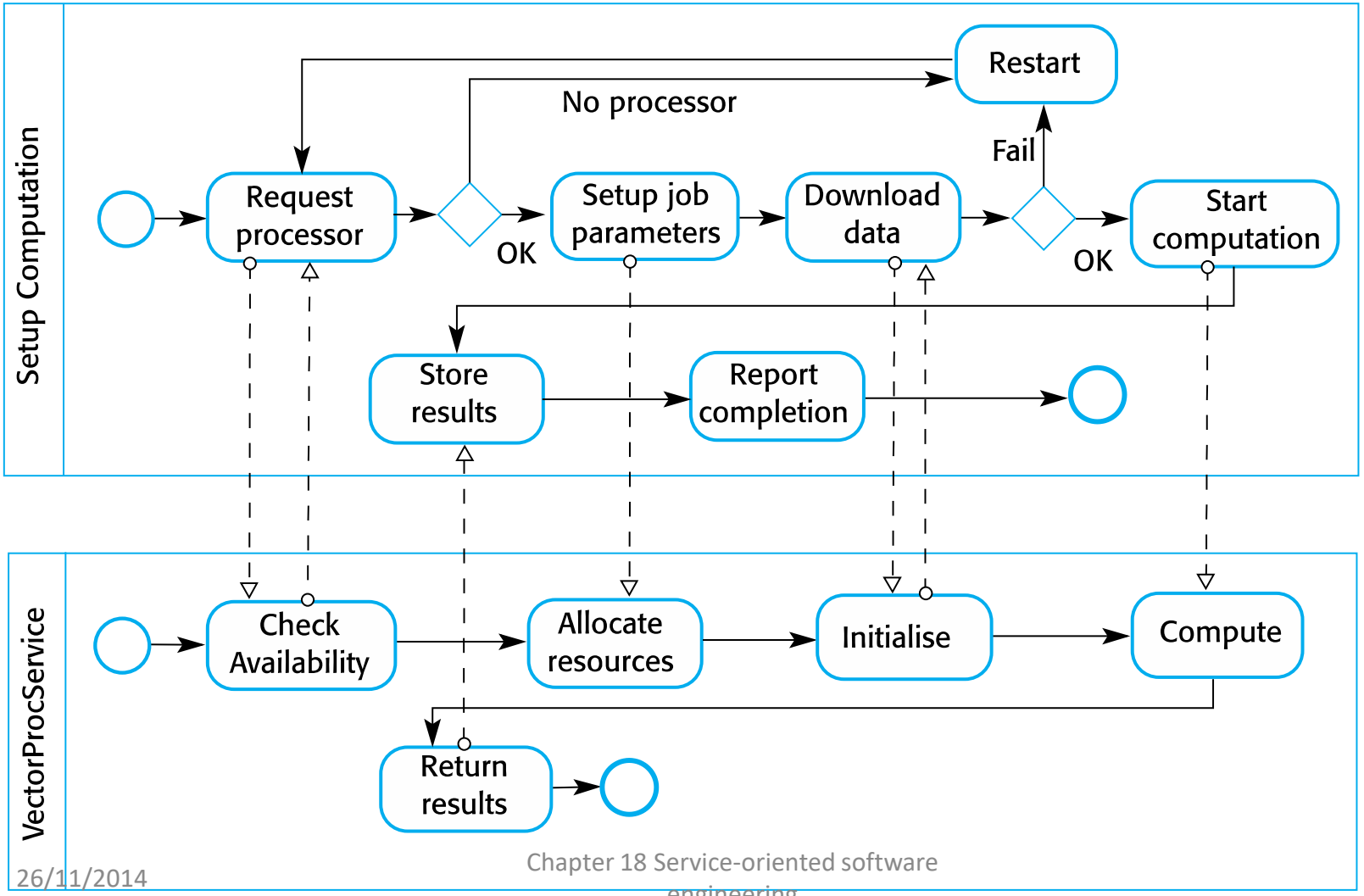


- WS-BPEL is an XML-standard for workflow specification. However, WS-BPEL descriptions are long and unreadable
- Graphical workflow notations, such as BPMN, are more readable and WS-BPEL can be generated from them
- In inter-organisational systems, separate workflows are created for each organisation and linked through message exchange.
- Workflows can be used with both SOAP-based and RESTful services.

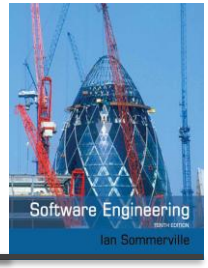
A fragment of a hotel booking workflow



Interacting workflows

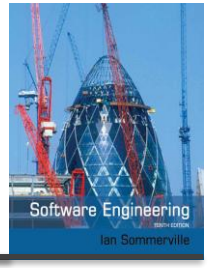


Testing service compositions



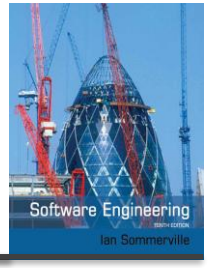
- Testing is intended to find defects and demonstrate that a system meets its functional and non-functional requirements.
- Service testing is difficult as (external) services are ‘black-boxes’. Testing techniques that rely on the program source code cannot be used.

Service testing problems



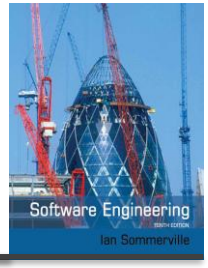
- External services may be modified by the service provider thus invalidating tests which have been completed.
- Dynamic binding means that the service used in an application may vary - the application tests are not, therefore, reliable.
- The non-functional behaviour of the service is unpredictable because it depends on load.
- If services have to be paid for as used, testing a service may be expensive.
- It may be difficult to invoke compensating actions in external services as these may rely on the failure of other services which cannot be simulated.

Key points



- Service-oriented architecture is an approach to software engineering where reusable, standardized services are the basic building blocks for application systems.
- Services may be implemented within a service-oriented architecture using a set of XML-based web service standards. These include standards for service communication, interface definition and service enactment in workflows.
- Alternatively, a RESTful architecture may be used which is based on resources and standard operations on these resources.
- A RESTful approach uses the http and https protocols for service communication and maps operations on the standard http verbs POST, GET, PUT and DELETE.

Key points



- Utility services provide general-purpose functionality; business services implement part of a business process; coordination services coordinate service execution.
- Service engineering involves identifying candidate services for implementation, defining service interfaces and implementing, testing and deploying services.
- The development of software using services involves composing and configuring services to create new composite services and systems.
- Graphical workflow languages, such as BPMN, may be used to describe a business process and the services used in that process.